Achyuta K Upadya

Software Engineer · Full-Stack & Platform · Bengaluru, India
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Backend engineer at HSBC Software Development — two years on Java and Spring Boot compliance platforms, and the sole DevOps engineer for a team running 40+ microservices. Took three platform-wide migrations end to end (CloudBees to open-source Jenkins, Helm and Kubernetes onboarding, NGINX to NGINX Plus), built a Spring Boot service from scratch to replace a 2018-era legacy system, and found and fixed two production access-control defects.

EXPERIENCE

HSBC Software Development Pvt Ltd · Bengaluru · Jul 2024 – Present

SOFTWARE ENGINEER — UNIFIED CASE MANAGEMENT · JAN 2025 – PRESENT

Java · Spring Boot · Kubernetes · Helm · Jenkins · NGINX Plus · HashiCorp Vault

- Sole DevOps engineer for a 40+ microservice compliance team — owning pipelines, Kubernetes deployments, secret management, and ingress.
- Built a Spring Boot proxy service handling JWT authentication and request routing, letting the case-management platform integrate securely with an external cloud-native system — designed, built, and shipped in ~4 weeks inside a legacy codebase.
- Found and fixed two production access-control defects in document handling: S3 objects reachable by users whose entitlements should have blocked them, and an Active Directory group check that could be bypassed — both closed with no data exposure.
- Led the move of 40+ services from CloudBees Jenkins to open-source Jenkins — ran the compatibility assessment and owned the full cut-over.
- Drove the Helm chart migration for 42 services after Jenkins deprecation, onboarding each onto the bank's internal Kubernetes deployment platform.
- Wrote the reusable deployment templates that wire HashiCorp Vault and Kubernetes into that platform, standardising how every service on the team receives its secrets.
- Migrated 42 services from NGINX to NGINX Plus — translated every config and validated traffic routing end to end.
- Partnered with the internal developer-experience (DevX) team on GitHub-based service templates covering repository setup, Jira, Jenkins pipelines, micro-frontend scaffolding, and JUnit generation — cutting new service bootstrap time by ~60%.
- Shipped a migration-readiness dashboard to production in 1.5 weeks (AI-assisted), giving product owners a live view of which case-management components were ready to be decommissioned ahead of the v2 rollout.
- Fixed a production defect in the request-for-information module, where question-and-answer content was being jumbled for investigators — patched with no downtime.

Software Engineer — goAML Compliance Platform · Jul 2024 – Jan 2025

Java · Spring Boot · React · GCP Cloud SQL · Liquibase · UNODC goAML

- Designed and built the Disclosure Service in Spring Boot from scratch, replacing a 2018-era UK legacy system — stood it up on its own GCP Managed Instance Group and wired it to the React front end.
- Extended the platform with a Global Suspicious Activity Report (SAR) module, adding Active Directory group-based routing and role-driven entitlement flows.
- Sole owner of PostgreSQL schema management on GCP Cloud SQL for the compliance platform — every schema change shipped as a Liquibase migration.
- Built reusable React components (dropdowns, screens, sections) for the platform's internal low-code UI framework, adopted across the project.
- Deputy DevOps owner for the goAML team — coordinated GCP deployments and release cycles.

Application Development Intern · Tele India Networks Pvt Ltd · Aug 2023 – Sep 2023

- Built an Android application in Android Studio for Datakosh, the company's private cloud product.

TECHNICAL SKILLS

Languages: Java, Go, Python, TypeScript / JavaScript, Swift
Backend & Frontend: Spring Boot, React, FastAPI, SwiftUI / SwiftData
Infrastructure & DevOps: GCP, Kubernetes, Helm, Docker, Jenkins, CI/CD, NGINX / NGINX Plus, HashiCorp Vault, Liquibase
Data & AI Tooling: PostgreSQL, Apache Kafka, Amazon S3, JWT / OAuth2, Git, Model Context Protocol (MCP)

PROJECTS

ashlar — Append-Only Commit Log · Go · github.com/achyuta0001/ashlar
The storage engine underneath a system like Kafka, on the standard library alone: segmented files, CRC-32C-checked records, and a fixed-width offset index that makes a read an array lookup. Recovers from a torn write or a lost index entry by replaying and truncating the incomplete tail — proven by a fuzz test that cuts the log at an arbitrary byte and asserts the survivors are always an intact prefix. 82% coverage under `-race`.

tripwyre — Project Intelligence CLI · Go · github.com/achyuta0001/tripwyre
Single-binary CLI that scans dependencies, config drift, and logs into one prioritised report. Pluggable scanner/adapter/reporter packages; adapters for npm, pip, cargo, dotenv, and JSON logs. 18 test files under GitHub Actions CI, ships as a reusable GitHub Action, runs fully offline.

Hive — Knowledge Compiler · Python · github.com/achyuta0001/hive
Ingests markdown, Notion, and Confluence docs into one canonical schema, clusters them by embedding similarity, and synthesises each topic into a merged wiki page — flagging contradictions instead of resolving them silently. Served to AI agents over MCP; content-hash diffing keeps LLM cost proportional to what changed.

Blister — iOS Die-Cast Collection Tracker · Swift · github.com/achyuta0001/blister
Native iOS app, Apple frameworks only — no backend, no third-party dependencies. On-device Vision OCR reads a collector card and auto-fills an entry against a bundled catalogue; RealityKit renders an orbitable 3D view. SwiftData persistence, ~170 commits.

obsidian-mcp — MCP Server · TypeScript · github.com/achyuta0001/obsidian-mcp
MCP server giving AI agents a per-project notes folder inside one Obsidian vault, so cross-project links and graph view keep working. Six tools; project inferred from the nearest `.git`. Tested end-to-end over JSON-RPC on stdio.

Kafka Fraud Detection Service · Java / Spring Boot · github.com/achyuta0001/Kafka-Fraud-Detection
Kafka consumer that scores transaction events against fraud rules — a high-value threshold and a velocity check on repeated transactions from one account — and publishes alerts downstream. Failed messages retry with backoff and then route to a dead-letter topic rather than being dropped.

Ollama Cross-Device Inference (macOS ↔ Windows) — 2026 — Ollama on a Windows machine exposed as a LAN inference endpoint for a MacBook, avoiding redundant model downloads.

Grok/X Context UI — 2026 — Web interface mapping live X (Twitter) context to Grok's API for conversational social data queries.

Hotel Review Sentiment Analysis — 2023 – 2024 — Full-stack mobile/web app running ML sentiment analysis across guest reviews, with per-property signal breakdowns.

Basic ERP UI — 2024 — React ERP frontend covering inventory, orders, and reporting.

EDUCATION & CERTIFICATIONS

B.E. Computer Science and Engineering — Nitte Meenakshi Institute of Technology, Bengaluru · Jan 2021 – May 2024

Anthropic ×2 (LinkedIn Learning, 2024) · Programming with JavaScript, Meta/Coursera (2023) · Introduction to Front-End Development, Meta/Coursera (2023) · Real World ML Applications — Dev & Deployment, Aqmenz Automation (2023) · NPTEL: Principles of Management, IIT Roorkee (2023) · NPTEL: Joy of Computing using Python, IIT Madras (2022)